MIL-P-24441/14A(SH)
15 May 1986
SUPERSEDING
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15 July 1980

MILITARY SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, EXTERIOR TOPCOAT, DARK GRAY, FORMULA 154R66-R_o = 3.6, TYPE II

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the paint described herein shall consist of this document and the latest issue of MIL-P-24441(SH).

FORMULA. This formula covers dark gray epoxy-polyamide paint designated Navy Formula 154R66, type II for camouflage, interior, or exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A

	Pounds
Thixotrope[1/]	-
Polyamide[2/]	20
Polyamide adduct[3/]	280
Magnesium silicate[4/]	400
Titanium dioxide[5/]	20
Lampblack[6/]	18
Paint thinner[7/]	179
2-ethoxy ethanol[8/]	133

Component B

	Pounds
Thixotrope[1/]	-
Epoxy resin[9/]	500
Magnesium silicate[4/]	250
2-butoxy ethanol[10/]	71
2-ethoxy ethanol[8/]	227

See footnotes on next page.

AMSC N/A FSC 8010
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QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. Components A and B shall be mixed 1:1 by volume for mixed components tests. Tests shall be performed in accordance with the general specification.

Requirements	Component A		Component B		Mixed component	
	Min	Max	Min	Max	Min	Max
Pigment content, percent	40.3	44.3	22.6	26.6	---	---
Volatiles, percent	34.0	38.0	26.2	30.2	---	---
Nonvolatile vehicle, percent	19.7	23.7	45.2	49.2	---	---
Water, percent	---	0.5	---	0.5	---	---
Coarse particles, percent	---	0.3	---	0.3	---	0.5
Consistency, grams	165	325	110	185	120	225
Pounds per gallon	10.2	10.6	10.2	10.6	10.0	10.6
Set-to-touch time, hours at 40°F	---	---	---	---	---	2
Set-to-touch time, hours at 73°F	---	---	---	---	---	2
Dry-hard, hours at 40°F	---	---	---	---	---	24
Dry-hard, hours at 73°F	---	---	---	---	---	6
Fineness of grind, NS	5	---	5	---	5	---
Flash point, °F	90	---	110	---	95	---
Titanium dioxide, percent of pigment	4	---	---	---	---	---
Pot life, hours at 73°F	---	---	---	---	5	---
Gloss, 60 degrees specular, percent	---	---	---	---	---	30
Sag resistance, mils	---	---	---	---	9	---
Color wet, Rw, percent	---	---	---	---	1.74	2.06
Contrast ratio, 3 mils dry film	---	---	---	---	0.98	---

1/ The amount and type of thixotrope shall be selected by the manufacturer to meet all requirements of the general specification and this specification sheet.
2/ GENAMID 2000, Henkel Corporation; UNIREZ 2810, Union Camp; AZAMIDE 600, AZ Products; ANCAMINE 507, Pacific Anchor Chemical Corp.; EPOTUF SF7791, Reichhold Chemical; TriChem 9200, Trimont Chemicals.
3/ VERSAMID 280B75, Henkel Corporation; UNIREZ 2180B75, Union Camp; AZAMIDE 680B75, AZ Products; ANCAMINE 700B75, Pacific Anchor Chemical Corp.; EPOTUF SF7792, Reichhold Chemical; TriChem 9280-B-75, Trimont Chemicals.
4/ Pfizer Talc MP 30-36; Cyprus Montana Talc, Mistron 400.
5/ DuPont Tipure R960; NL Chemicals, TX 2101; Kerr-McGee, Tronox Tr 822.
6/ Pfizer M1011.
7/ Chevron 350H, VARSOL-18, AMSCO Mineral Spirits.

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8/ 99 percent pure by gas chromatography, density 0.93, refractive index (68°F) = 1.407, IR spectrum characteristic of compound.
9/ EPON 828, Shell Chemical Co.; ARALDITE 6010, Ciba-Geigy; DER 331, Dow Chemical Co.; AZEPOXY 128, AZ Products; EPO-TUF 37-140, Reichhold Chemical; EPI-REZ 510, Celanese; TriChem 728, Trimont Chemicals.
10/ 99 percent pure by gas chromatography, density 0.90, refractive index (68°F) = 1.418, IR spectrum characteristic of compound.

Use of alternate ingredients in this formula must have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that paint made using the ingredient will conform to all requirements of this specification.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
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